O12 Phase 1C - ready for owner review

18 October 2026. Targeted corrections complete; owner approval pending. Phase 2 remains unauthorised.

Exact changes

- Foundation 7b and Higher 7b: indicative marking guidance now explicitly credits comparing several plant tissue types under matched conditions, repeating within each type and comparing zero-change concentrations. The top band requires developed reasoning addressing generalisation. Repeats of one tissue improve reliability but cannot test the universal claim. Agreement among sampled tissues cannot prove it for every plant tissue.

- Both six-mark rubrics retain holistic best-fit levels, 0-6 marks, equivalent valid science and alternative approaches (including a justified counterexample). No mandatory wording or keyword checklist.

- Foundation model response now describes the tissue comparison and its implications. Higher model remains unchanged because it already calls for separate tissue tests; the expanded rubric supplies the matched-condition detail.

- Higher 5b now asks “how many times as large as Q’s it is”. Its answer remains 12:1 and four times Q.

Verification

Each tier retains 40 marks, seven groups, 24 parts, AO1/AO2/AO3 allocation 16/16/8 and the 11 selected-response / 29 reviewed-mark split. Questions, keys, numerical inputs and all 22 SVGs and 22 PNG previews are unchanged apart from the Higher 5b wording. Foundation question PDF is byte-identical. Targeted catalogue/text diffs and regenerated PDF contents were checked for alignment; calculations and table/graph coordinates were recomputed. Revised PDF pages were rendered and visually checked.

Calculations: F2c 40 μm; H2c 3 × 10^-6 m; F5b 3:1; H5b 12:1 and factor 4; F7a +10%; H7a -8.8%. Graph zero estimates: Foundation 0.40, Higher approximately 0.318 mol/dm³ (accept 0.31-0.33).

Only copied O12 approval materials were edited. O1-O11 and application files were not edited. Existing application-test evidence remains historical; no new app tests are claimed. No Phase 2 work, commit, publication, paid service or dependency purchase.

Remaining limits

Learner timing/difficulty trials, live behaviour, physical iPhone checks and official-equivalence validation remain outstanding. AO totals are the existing design allocation, not certified AQA equivalence.

The Phase1B-Evidence folder contains historical reports, not fresh Phase 1C results. Phase1C-checks.json and Phase1C-diff.txt record this revision. Stop here for owner approval.